
DL for ENG
Designed by Remus Teodorescu 2026

LEARNING OBJECTIVES

What You Will Be Able To Do

THE GOALS, IN PLAIN WORDS

- **interrogate** — any AI result you are shown, using four questions that always apply
- **recognise** — the same three ingredients behind results in six unrelated fields
- **distinguish** — a product from an instrument, and a demonstration from a deployment
- **judge** — when a surrogate model is the right answer, and when it is a shortcut
- **state** — why publication bias makes this lecture a biased sample, including this slide
- **arrive at L4** — with the framing done, and ready for the technical half

WHAT TODAY IS NOT

not	a survey to memorise
not	a claim that all of this works
not	technical — L4 starts that

THE ONE HABIT TO LEAVE WITH

Four questions, asked of every result: what is the input, what is the loss, where did the data come from, and what happens when it is wrong. They are on the next slide and they are how every report in this course is marked.

Four Questions to Ask of Every Example

THE HABIT THIS LECTURE IS TRYING TO BUILD

- **twenty results in forty minutes** — and that is exactly when engineers stop asking questions
- **what is the input** — how many variables, on what grid, at what cadence?
- **what is the loss** — what single number was minimised, and did anyone want it?
- **where did the data come from** — and who paid? This decides whether it transfers.
- **what happens when it is wrong** — a bad film recommendation and a bad plasma action differ

THE FOUR, SHORT

input	what exactly goes in?
loss	what number was minimised?
data	where from, at what cost?
failure	what happens when it is wrong?

WHY THIS MATTERS MORE THAN THE EXAMPLES

In twelve months you will have forgotten most of what is on the mosaic. If you keep the four questions, you will be able to evaluate results that do not exist yet — which is the only durable skill this lecture can give you.

Twenty Domains, One Method

VISION classification, detection, segmentation	**LANGUAGE** translation, summarisation, code	**SPEECH** recognition, synthesis, diarisation	**GAMES** Go, chess, StarCraft, poker	**RECOMMENDATION** the commercial engine of the field
PROTEIN STRUCTURE AlphaFold — a fifty-year open problem	**WEATHER** GraphCast — a GNN on a global mesh	**MATERIALS** candidate crystal discovery at scale	**FUSION** tokamak plasma shape control	**ASTRONOMY** transient detection, lensing, surveys
SURROGATES replace a slow solver in a design loop	**MAINTENANCE** anomaly detection on plant sensors	**INSPECTION** weld, casting and surface defects	**DESIGN SEARCH** topology and layout optimisation	**CONTROL** learned policies on real hardware
HEAT TRANSFER L8 — steady and transient, on real geometry	**FLUID FLOW** L9 — Burgers, channel flow, turbulence	**ELECTROCHEMISTRY** L10 — batteries and solid oxide cells	**AUTONOMOUS ROBOTS** L11 — a JetRacer you will actually drive	**POWER GRID** L12 — state estimation and stability

● the classics ● science ● engineering ● this course

Three Tasks, Increasing in Difficulty

WHAT CHANGES BETWEEN THEM IS THE OUTPUT

- **classification** — one label for the image. Is there a crack?
- **detection** — a label and a box for each object. Where are the cracks?
- **segmentation** — a label for every pixel. Usually what engineers actually want.
- **the network barely changes** — only the loss and the output layer

THE FOUR QUESTIONS, ANSWERED

input	a grid of pixels, 3 channels
loss	cross-entropy, per image or per pixel
data	hand-labelled — the expensive part
failure	confident label on an unseen defect

THE ENGINEERING CATCH

Segmentation needs a human to outline every defect in every training image. On a real inspection project that labelling effort routinely exceeds the modelling effort by a factor of five, and it is the line item that kills proposals.

Where It Already Earns Its Keep

SURFACE AND WELD INSPECTION

In production on lines that run faster than a human inspector can watch. The economics work because the camera is cheap and the defect rate is low but expensive.

DIMENSIONAL METROLOGY

Measuring from images rather than with a probe. Faster, contactless, and accurate enough for a great many tolerances — though not for all of them.

MEDICAL IMAGING

Screening support in radiology and pathology, deployed and regulated. Note the word support: the approved systems flag and prioritise, they do not decide.

WHY VISION WENT FIRST

Three reasons, and they are worth naming because they tell you which of your own problems are ready.

The input has strong local structure, which convolution exploits directly — L5.1.

Labels are cheap relative to other fields: a human can label an image in seconds, where labelling a fatigue test takes a week of rig time.

And the failure is usually visible. When a vision model is wrong, a person can look at the image and see that it was wrong — which is not true of a model that predicts a stress field.

Language, and Why It Drove Everything Else

THE TASKS, BRIEFLY

- **translation** — the benchmark that motivated the transformer
- **then everything else** — summarising, question answering, code
- **why it matters here** — language is where the field learned to scale
- **self-supervision** — hide a word, predict it. No human labels at all.
- **the engineering analogue** — L6.2, and it is the most valuable export

THE FOUR QUESTIONS

input	a sequence of tokens
loss	predict the next token
data	scraped text — legally contested
failure	fluent, confident, false

SELF-SUPERVISION, THE EXPORTABLE IDEA

Hide part of the input and predict it. No labels required. For an engineer sitting on a year of unlabelled sensor data and no fault annotations, that is not a curiosity — it is the most promising thing on this slide.

Large Language Models, in Four Sentences

WHAT THEY ARE

A transformer trained to predict the next token over an enormous corpus, then adjusted afterwards to be useful and to follow instructions — a process called post-training, which is the subject of L6.2. That is the whole architecture.

WHAT THEY ARE GOOD FOR, IN ENGINEERING

Reading and summarising documentation, standards and reports — a real time saving, and low risk because you can check the source.

Writing and explaining code, which is how most engineers now encounter them.

Extracting structure from unstructured maintenance logs and free-text fault reports. This one is underrated and quietly valuable.

In all three cases the common feature is that a human verifies the output cheaply.

WHAT THEY ARE NOT

They are not calculators, and they are not simulators. A language model asked for a heat transfer coefficient will produce a plausible number, and plausible is precisely the problem.

They have no model of the physical system. Nothing in the training objective rewards being right about a quantity — only about the next token.

This course is not about them, and they appear nowhere in L7 to L12. I name them because you will be asked at a family dinner what you are studying.

The verification cost is the whole test: use them where checking the answer is cheaper than producing it.

Speech, Games, and the Route to Real Control

TWO MORE, AND WHY THE SECOND MATTERS HERE

- **speech** — transformed around 2012, and now solved well enough to be invisible
- **AlphaGo, 2016** — more board positions than atoms in the observable universe
- **then it moved to physical systems** — chip floorplanning, data-centre cooling, plasma
- **the equivalence** — a game is a simulator with a score. So is a plant model.
- **which is why** — L6.2 teaches RL and L11 puts a policy on a car

GAME TO PLANT, IN FOUR STEPS

2016	AlphaGo — a board, and a win/loss
2017	self-play, no human games needed
2020	data-centre cooling in production
2022	tokamak plasma control, on hardware
your turn	L11, on a car that can crash

THE STEP THAT IS ACTUALLY HARD

Not the algorithm — the simulator. A policy learned in simulation and deployed on hardware meets every modelling error at once. That gap has a name, the sim-to-real gap, and L11 is where you will feel it.

The Turn: From Products to Instruments

WHY THE NEXT FIVE SLIDES ARE THE POINT OF THIS LECTURE

- **everything so far** — was a product. Real work, and it funded the field.
- **the last five years** — produced something different
- **the network as instrument** — not a faster answer, but a finding that was not possible before
- **the 2024 Nobel in Chemistry** — went to one of them
- **so keep the four questions** — these are the results most likely to be over-read

FIVE RESULTS, AND WHAT EACH BROKE

proteins	a 50-year prediction problem
weather	a physics monopoly on forecasting
materials	the rate of candidate discovery
fusion	control design by hand-tuning
astronomy	the limit on survey throughput

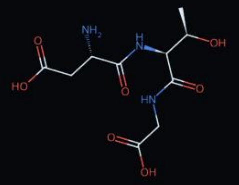
Notice what they have in common. In every case the physics was already understood; what was missing was a way to search a space too large to enumerate. That is a very specific kind of problem, and recognising it is a skill.

AlphaFold: A Fifty-Year Problem, Closed

THE PROBLEM, AND WHY IT WAS HARD

- **the problem** — a chain of amino acids folds into a shape, and the shape is the function
- **open for fifty years** — not for want of physics. The forces are known.
- **the difficulty** — the conformation space is astronomical and the energy landscape is rough
- **AlphaFold 2, 2021** — accuracy comparable to experiment, for a large fraction of proteins

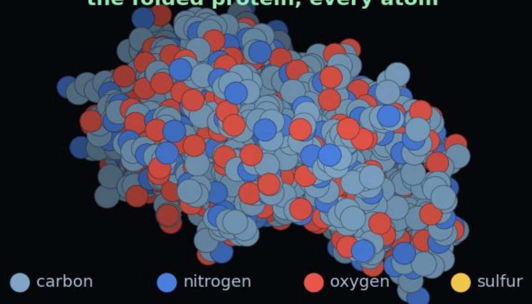
three amino acids, joined



PQVTLWQRPLVTIKIGGQLKEALLDT

the sequence — cheap to read

the folded protein, every atom



HIV-1 protease — 1,500 atoms

THE FOUR QUESTIONS

input	amino acid sequence, plus relatives
loss	distance and angle error vs. structure
data	decades of public crystallography
failure	confident on disordered regions

THE LESSON FOR AN ENGINEER

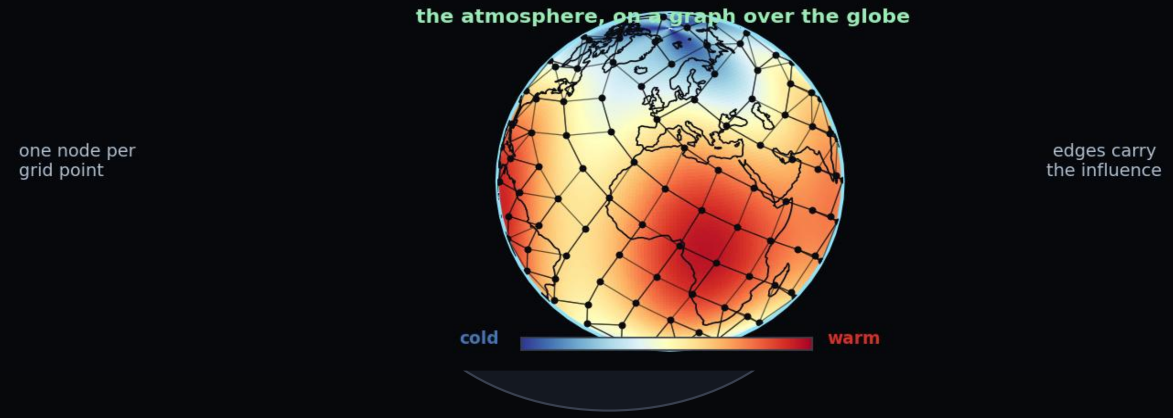
Fifty years of publicly deposited experimental structures is what made this possible. The model is remarkable; the dataset was the precondition. When you read a result like this, ask who spent fifty years collecting the data.

Jumper et al. (2021), Nature 596

GraphCast: A Graph Network on a Planet

THE ONE THAT SHOULD MOST INTEREST THIS ROOM

- **numerical weather prediction** — the great success of computational physics
- **GraphCast, Science 2023** — learned medium-range forecasting from reanalysis data
- **it outperformed** — the leading physical model on most variables, in under a minute
- **the architecture** — a graph network on a mesh over the globe. You build one in L5.1.



THE FOUR QUESTIONS

input	gridded atmospheric state, 2 times
loss	weighted error at future lead times
data	ERA5 reanalysis — decades, public
failure	unprecedented events, poorly

WHAT IT DOES NOT DO

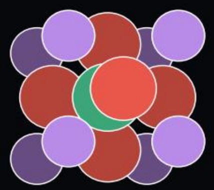
It does not replace the physics. It was trained on reanalysis, which is itself the output of a physical model fused with observations. Remove the physics-based pipeline and there is no training data. Say that when someone tells you weather forecasting has been solved by AI.

Materials, and Looking at the Sky

TWO SEARCHES, ONE SHAPE

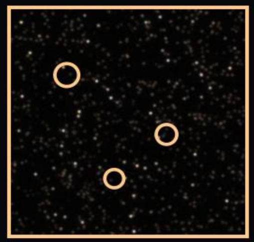
- **materials** — is this crystal stable? Screen millions cheaply, then compute the survivors.
- **GNoME, 2023** — an order-of-magnitude change in the size of the candidate pool
- **astronomy** — more images each night than any group of astronomers can inspect
- **in particle physics** — learned triggers decide what is even worth recording, in real time

a candidate crystal

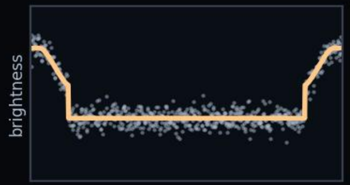


millions screened cheaply,
thousands worth making

one field of sky, and one transit



every point is a galaxy
circled: worth a second look



a 2% dip, buried in noise
millions of stars, every night

WHY IT WORKS HERE

the filter	cheap model first, expensive method after
the payoff	the expensive method runs less often

THE HONEST USE

This is the most defensible use of deep learning in science: not replacing the physics, but deciding where to spend it.

AND THE CAUTION

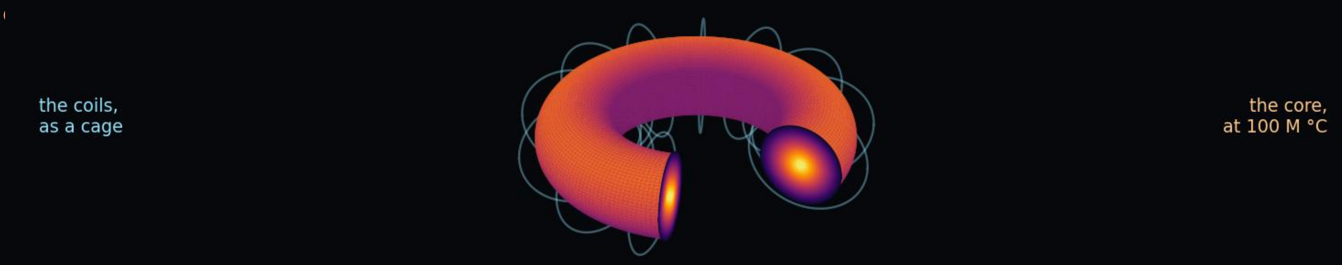
A candidate is a hypothesis. GNoME predicted structures; synthesis is another matter.

Controlling a Plasma at a Hundred Million Degrees

A CONTROL PROBLEM, SOLVED THE HARD WAY

- **a tokamak** — holds plasma off the wall with magnetic fields from a set of coils
- **conventionally** — each configuration needs its own controller, tuned over weeks
- **DeepMind and EPFL** — trained one RL policy in simulation, then ran it on the real TCV machine
- **it held shapes** — that had not been demonstrated before

a tokamak, cut open — the plasma is a ring



held clear of the wall by the field alone — and the balance drifts in milliseconds

THE FOUR QUESTIONS

input	magnetic measurements, real time
loss	reward on shape and position error
data	generated in a physics simulator
failure	the plasma touches the wall

WHY THIS RESULT MATTERS HERE

Almost no measured data. A trusted physical simulator. A control problem where failure is expensive and exploration is forbidden. That is a very common engineering situation, and it is precisely the situation L11 puts you in — with a cheaper car.

Degrave et al. (2022), Nature 602

Surrogate Models: The One You Will Actually Build

THE MOST COMMON ENGINEERING USE, BY SOME MARGIN

- **the situation** — a solver you trust, four hours a run, and a loop needing ten thousand
- **the surrogate** — run it a few hundred times, train a network, evaluate that instead
- **microseconds not hours** — and this is the most common industrial use of deep learning
- **not new** — response surfaces date to the 1950s

$$f_\theta(x) \approx \mathrm{Solver}(x) \qquad t_f \ll t_{\mathrm{Solver}}$$

- **what changed** — dimensionality. A network takes a whole geometry or field as input.

The surrogate contract: close enough, and very much faster.

SOLVER		SAMPLES		TRAIN		EVALUATE
4 hours each	→	a few hundred runs	→	once, overnight	→	microseconds, for ever

Pay once, then evaluate ten thousand designs. That trade is the whole idea.

THE FOUR QUESTIONS

input	design variables, or a whole field
loss	error against the solver's output
data	your own solver — you make it
failure	silently wrong off the sample set

THE TRAP, AND IT IS A COMMON ONE

A surrogate is only valid inside the region you sampled. Optimisers are attracted to the edges of that region, because that is where the surrogate's errors look like improvements. Your optimiser will find the surrogate's mistakes for you, reliably.

Predictive Maintenance, and the Data You Already Have

THE PROBLEM IS NOT THE MODEL

- **the pitch** — every plant already logs sensor data. Learn normal, flag departures.
- **the structure** — much data, almost no labels, because the faults are rare
- **the honest difficulty** — base rates, not the model
- **no architecture fixes this** — it is a statistics problem

1% false alarms on hourly data, against a fault every three years

the detector is not broken — the base rate is,
and no architecture fixes it

WHY THESE PROJECTS STALL

labels	the faults were never annotated
base rate	rare events, many false alarms
drift	the plant changes; the model does not
trust	one bad alarm and it is ignored
ownership	nobody maintains the model

Every item on that list is organisational rather than mathematical. If you take on one of these projects, budget your effort accordingly — and notice that none of the five is taught in a deep learning course, including this one.

Design Search — and a Word About “Generative”

A TERMINOLOGY TRAP WORTH CLEARING UP NOW

“Generative design” in CAD is topology optimisation — a classical, physics-based method that long predates deep learning. It is not generative AI. The words collided; the methods did not.

WHAT IS CLASSICAL HERE

Topology optimisation: remove material where stress is low, iterate, converge on a shape. Physics all the way down, and it works.

Gradient-based shape optimisation with an adjoint solver — the aerodynamicist's tool, and still the best available when you can afford it.

Neither of these needs a neural network, and neither is improved by adding one.

WHERE LEARNING ACTUALLY HELPS

Speeding up the inner loop with a surrogate, as on slide 14 — this is the real contribution.

Learning a compact description of a family of shapes, so the search happens in ten dimensions instead of ten thousand.

Predicting a good starting point from previous designs, so the classical optimiser converges in a fraction of the iterations.

In all three the classical method is still doing the engineering. The network makes it affordable.

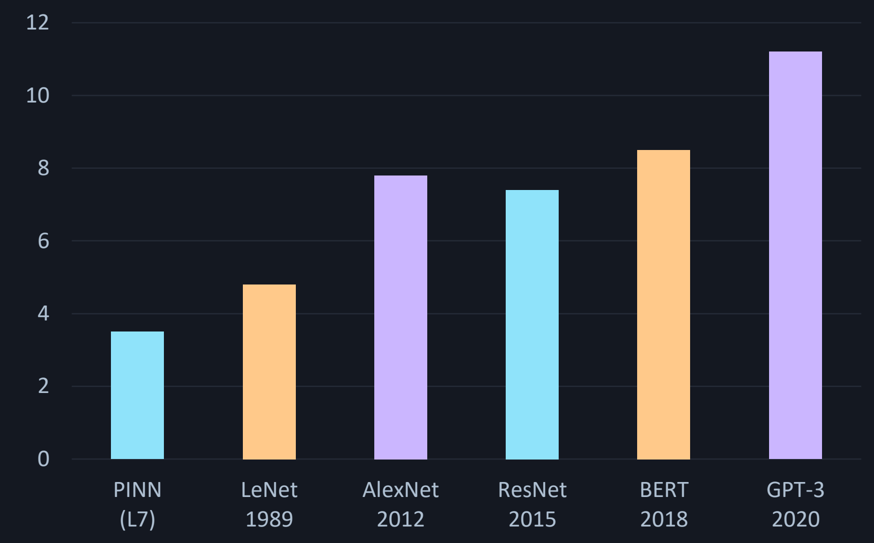
A recurring pattern: the learned component rarely replaces the physics — it removes the cost that made the physics impractical.

What It Costs

THE NUMBER NOBODY PUTS ON THE SLIDE

- **training compute** — up by orders of magnitude in a decade, and the energy with it
- **almost entirely irrelevant to you** — every model in this course trains on a CPU in minutes
- **the PINNs in L7** — minutes on a laptop, and that is the point
- **why show it** — you will be asked about energy, and the gap is the most misunderstood thing

PARAMETERS, LOG SCALE, ILLUSTRATIVE



Order-of-magnitude figures for teaching, not a citable table. The left-hand bar is the kind of model you will actually build in this course.

Shipped, Assisting, and Still Hard

WHAT ACTUALLY HAPPENED

- **driving** — supervised autonomy is on public roads in several countries, at scale
- **radiology** — AI reads run alongside the radiologist. Assisting, not replacing.
- **drug discovery** — MAMMAL, 2026: one model, nine of eleven benchmarks at state of the art
- **the pattern** — arrival is quieter than the promise, and later, and narrower

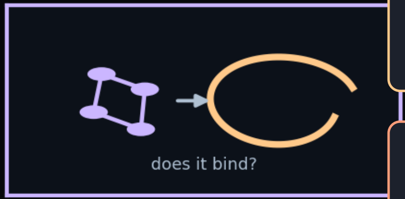
AUTONOMY

on public roads, supervised

RETINAL SCREENING

alongside the clinician

DRUG DISCOVERY



MAMMAL, 2026

all three shipped. None of them replaced anybody.

THE BREAKTHROUGH TO KNOW

MAMMAL	proteins, molecules and omics, one model
trained on	2 billion biological samples
and	beat AlphaFold 3 on 5 of 7 antibody targets

WHAT IS STILL HARD

Unsupervised driving anywhere, in any weather. Diagnosis without a clinician. Both are narrower problems than the headlines implied.

AND THE STRUCTURAL BIAS

Abandoned projects are not published. You cannot read the failures.

Shoshan et al. 2026, npj Drug Discovery

How to Read a Claim

FIVE QUESTIONS, ASKED IN THIS ORDER

- **against what** — a real baseline, or a straw man nobody uses?
- **on whose data** — the authors' own, or a set someone else chose?
- **what breaks it** — did they report a failure mode, or only successes?
- **can it be repeated** — is the code public, and the seed stated?
- **who paid** — and what would they have done with a negative result?

APPLY THEM TO THIS LECTURE

baseline	mostly stated — check GraphCast's
failure modes	one slide out of thirty-one
who paid	three of these came from one company

WHY THIS IS THE SKILL, NOT THE ARCHITECTURES

Architectures change every eighteen months and are freely available. These five questions do not change, and they are how every report in this course is marked. Ask them of your own work first — it is cheaper than having someone else ask.

THIS COURSE

The Five You Will Build

L8 · HEAT TRANSFER

Steady conduction on a plate with a hole, then the transient problem — and then the inverse problem, recovering a diffusivity you were not given.

L9 · FLUID FLOW

Burgers' equation in two dimensions, then channel flow. The lecture where you learn to balance competing terms in a loss.

L10 · ELECTROCHEMISTRY

Ionic diffusion and charge conservation in a battery, then a solid oxide cell. The first genuinely coupled multiphysics problem.

L11 · AUTONOMOUS ROBOTS

Six JetRacer cars. Camera navigation, then vehicle dynamics and energy — checked against an instrument rather than an exact solution.

L12 · POWER GRID

State estimation from partial measurement on a network, then stability prediction. Most of the machine is unmeasured, permanently.

Five different physics. One method, unchanged from L7 slide one: write the physics as a residual, build in what you know exactly, let the optimiser find the rest.

THE PATTERN

What Every Example Today Had in Common

TWENTY DOMAINS, ONE EXPRESSION

- vision, language, proteins, weather, plasma, surrogates
- **every one** — a parameterised function, a loss, an optimiser
- **what differs** — not the mathematics. What goes into the loss, and where the data came from.
- **which is why** — this course spends its second half on the loss, not on architectures

$$\hat{\theta} = \arg\min_{\theta} \frac{1}{N} \sum_{k=1}^{N} \ell(f_\theta(x_k),\ y_k)$$

The supervised form. In L7 the second argument disappears and a differential operator takes its place.

AND WHAT DIFFERED

vision	labels from people, in seconds each
language	no labels — hide and predict
proteins	fifty years of crystallography
weather	decades of physics-based reanalysis
fusion	a simulator, and no real exploration
L7 – L12	no data at all — a residual instead

Read that column downwards. It is a list of increasingly clever answers to one question: where do you get supervision when nobody will label anything for you?

YOUR TURN, NOW

Your Own Model, on the Spectrum

THE EXERCISE I ASKED YOU TO PREPARE

You were asked to bring one model from your own work. In pairs, for five minutes: say what it is, place it on the modelling spectrum, and name the one thing you would need to move it one step towards physics. Then we take four of them.

PROMPTS, IF YOUR PARTNER IS STUCK

what is the input?	and how many of them do you have?
what did you minimise?	and was it what you wanted?
where did the data come from?	who paid, and how long did it take?
what happens when it is wrong?	and would you notice?

WHAT USUALLY COMES OUT OF THIS

Most models in the room turn out to be further towards the data end than their owner expected — usually because a physical form was chosen for convenience and then fitted freely.

The second common finding is that nobody can state the loss. It was whatever the software minimised by default.

Neither is a criticism. Both are the reason this course exists, and both are much easier to fix once named.

What Changes in Lecture Four

THE END OF THE HAND-WAVING

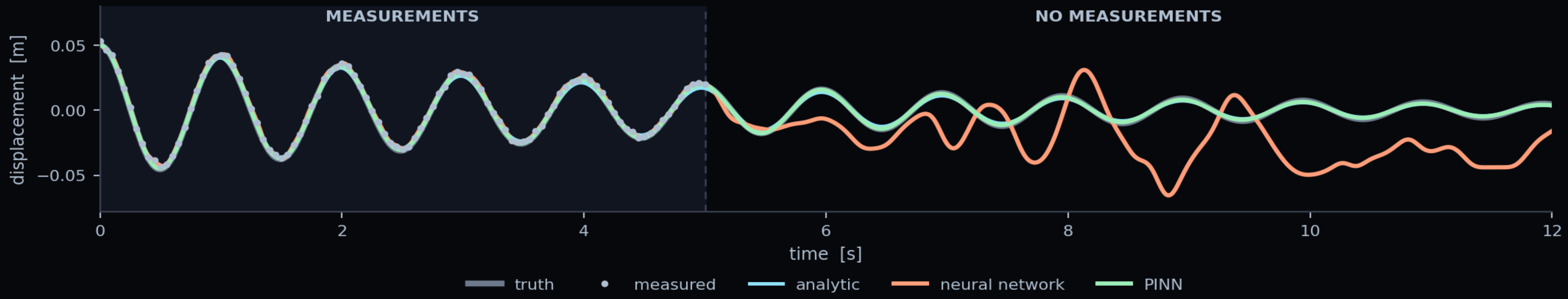
- **two lectures of framing** — is enough. From L4 it is technical, continuously.
- **L4** — the network from nothing: linear model, perceptron, its famous failure, depth
- **L5** — the architectures that matter: convolutional, graph, sequence, attention
- **L6** — training, and what comes after. Four slides in L6.1 make L7 possible.
- **the framing stays** — the four questions are how every report is marked

WHAT YOU NOW HAVE

vocabulary	AI, ML, DL, and the three kinds
the three ingredients	function, loss, optimiser
the spectrum	physics ↔ hybrid ↔ data
the four questions	input, loss, data, failure
a sense of scale	what is real and what is claimed

That is genuinely enough to read most of the literature critically, which is more than two lectures usually buy. Everything from here is depth.

The Four Questions, on Your Own Four Models



WHAT YOU WILL BUILD

- **one system** — a mass on a spring, released and left to ring
- **four models** — analytic, regression, network, PINN. The same 9,473 parameters twice.
- **the surprise** — the orange curve fits the measurements better than they are correct
- **the report** — asks the four questions from slide 3 about each of your own models

ERROR OUTSIDE THE DATA

analytic	1.34 mm
regression	3.03 mm
network	30.26 mm
PINN	0.59 mm

THE QUESTION THAT CARRIES THE MARKS

All four fit the data. Which would you put in a design report, and why? The marks are for the reasoning, not the error.

RESPONSIBILITY

Ethics as an Engineering Constraint

NOT A SEPARATE SUBJECT

- **you already work under obligation** — a structural engineer signs off a design and owns it
- **the model is not the responsible party** — that temptation is the thing to resist
- **regime shift** — trained on one operating condition, deployed in another
- **the confident wrong answer** — wrong in a way nobody can detect
- **the automation paradox** — the operator loses the ability to notice the failure
- **the version that will affect you** — can you say what your model does not know?

WHERE THIS RETURNS

L6.2	what a validation set does not prove
L11	a model on hardware that can crash
L12.1	an estimate is not a measurement
every report	state what you assumed

THE HABIT THIS COURSE ENFORCES

Every report asks you to separate what you measured from what you inferred, and to say what would have to be true for your conclusion to hold. Not ethics bolted on — professional practice, written down.

SUMMARY

Key Takeaways

1. **Ask four questions, every time**
   Input, loss, data, failure. They outlast every example on the mosaic.

2. **The science results are the real evidence**
   Proteins, weather, fusion — cases where a network was the instrument that made a finding possible.

3. **The physics rarely gets replaced**
   It gets made affordable. GraphCast needs reanalysis; the surrogate needs the solver; the plasma policy needs the simulator.

4. **The failures came from narrow proxies**
   Driving, radiology, clinical support. The model did what it was asked; the loss did not encode the job.

5. **Supervision is the scarce resource**
   Labels, self-supervision, archives, simulators — and in L7, a differential equation instead.

Next — L4.1: the end of the framing, and the beginning of the machinery.

OVER TO YOU

Questions

SIX TO TEST WHETHER TODAY LANDED

- **what would you ask** — of the next impressive result somebody shows you?
- **how would you tell** — a demonstration from something you could deploy?
- **what makes a problem** — a good fit for this, and what rules it out?
- **where does the data come from** — in your field, and who paid for it?
- **what does the published record** — leave out, and how would you find it?
- **your own work** — which of the four questions can you not yet answer about it?

BEFORE L4

read	UDL ch. 1 — twenty minutes
run	the Ex03 notebook, end to end
bring	one model from your own work

AND ONE WITH NO EASY ANSWER

You have been shown twenty successes. What would a lecture of twenty failures have taught you instead, and why can nobody give it?

WHERE THE HOUR LANDED

The architectures are freely available. The loss is where the engineering is, and that is the second half of this course.

SOURCES

References and Further Reading

THE PRIMARY RESULTS SHOWN TODAY

Jumper et al. (2021), Highly accurate protein structure prediction with AlphaFold, Nature 596 — open access.

Lam et al. (2023), Learning skillful medium-range global weather forecasting, Science 382 — the GraphCast paper.

Degrave et al. (2022), Magnetic control of tokamak plasmas through deep reinforcement learning, Nature 602 — the TCV work, with EPFL.

Merchant et al. (2023), Scaling deep learning for materials discovery, Nature 624.

THE TEXTBOOKS

Prince, Understanding Deep Learning — ch. 1 for the taxonomy, ch. 10.5 for vision applications, ch. 12.5–12.8 for language, ch. 19 for reinforcement learning.

Goodfellow, Bengio & Courville, Deep Learning — ch. 12, Applications, is dated but its framing of what makes a task tractable has held up well.

READ IN THIS ORDER

if you read one	the GraphCast paper — it is a GNN, and L5.1
if you read two	the tokamak paper — it is L11, at scale
before L4.1	UDL ch. 2 and 3
for the exercise	the run guide in Ex_03
for the argument	the honest-claim slide, again

All four primary papers are open access. The DeepMind blog posts are readable summaries but they are marketing — read the papers for the caveats, which are in the papers and not in the posts.

BEFORE YOU GO

One Thing Worth Being Careful About

THE FAILURE MODE OF A LECTURE LIKE THIS ONE

- **twenty domains, and almost all worked** — that is a biased sample
- **the bias is structural** — failures are not published, so you cannot read them
- **slide 18 is in the middle** — so it does not read as a disclaimer
- **the right response** — is not scepticism. The method works.
- **it is to ask the four questions** — every time, including of this lecture

SURVIVORSHIP, IN ONE LINE

You cannot see the projects that failed, so you will systematically overestimate how often this works.

This is not specific to machine learning. It is how every applied field looks from the outside, and engineers are usually good at spotting it — right up until the demonstration is impressive.

The four questions are a defence against your own enthusiasm, which is the only kind of defence that is ever needed.

Nothing in this course is harder than that, and nothing in it matters more.

NEXT

L4.1 — From Perceptron to Neural Network

Two lectures of framing are done. Next time we build the thing itself, starting from a straight line and ending with a proof that a shallow network can approximate anything — and an explanation of why that proof matters less than it sounds.

BEFORE THE NEXT SESSION

read	UDL ch. 2 and 3 — about an hour
run	Ex_03, the one notebook
bring	your extrapolation plot from Ex_03
think about	why the network extrapolated worst

Bring the plot. L4.1 opens by explaining what you will have seen, and the explanation is much more convincing if you have already been surprised by it once.